SAGAR SHUKLA

Geospatial Engineer · Remote Sensing & GeoAI

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PROFESSIONAL SUMMARY

Computer Science graduate with 2+ years of hands-on technical experience developing automated satellite data pipelines, raster processing workflows, and geospatial AI models. Proficient in Python-based Earth Observation (EO) toolchains (GDAL/Rasterio, GeoPandas, Google Earth Engine, xarray, STAC), spatial data management, and custom deep learning architectures (U-Net, TerraTorch). Experienced in deploying cloud-backed pipelines, writing reproducible GIS analyses, and processing multi-sensor Earth observation data (Sentinel-1 SAR, Sentinel-2, CHIRPS) for real-world environmental and remote sensing applications.

TECHNICAL SKILLS

- **Geospatial & Remote Sensing:** Google Earth Engine (Python API), GDAL, Rasterio, GeoPandas, Fiona, Shapely, PyProj, xarray, rioxarray, QGIS, ArcGIS Pro, SNAP, pystac-client (STAC API), Cloud-Optimized GeoTIFFs (COG), CRS Reprojection, Sentinel-1 (SAR), Sentinel-2 (MSI), CHIRPS, Spectral Indices (NDWI, NDVI)
- **Spatial Databases & Cloud:** PostgreSQL / PostGIS (spatial queries, spatial indexing, geometry operations), SQL, GCP (Cloud SQL, Cloud Storage), Docker, AWS (S3 object storage basics)
- **Machine Learning & GeoAI:** PyTorch, PyTorch Lightning, TerraTorch, U-Net (semantic segmentation), OpenCV, scikit-learn, CNNs, YOLO, transfer learning, IoU/F1 evaluation
- **Languages & Tools:** Python, SQL, Git / GitHub, Streamlit, Linux/Bash, REST APIs, Jupyter, Pandas, NumPy, Matplotlib

CORE GEOSPATIAL & ENGINEERING PROJECTS

Automated EO Pipeline & Water-Change Analytics — *Sentinel-2, GEE, Rasterio, CHIRPS* **github.com/Saanctity/kolar-dam-water-change-detection**

- **Automated Acquisition:** Engineered an end-to-end data acquisition and raster analysis pipeline to monitor surface water dynamics across multiple years using multi-temporal Sentinel-2 imagery.
- **Cloud Masking & Compositing:** Automated cloud-masking, median compositing, and NDWI thresholding via Google Earth Engine Python API and STAC queries (pystac-client).
- **Reprojection & Raster Math:** Built local raster processing routines using Rasterio, xarray, and PyProj, executing accurate geometric transformations and surface area calculations by reprojecting datasets from EPSG:4326 to UTM Zone 43N.
- **Meteorological Correlation:** Integrated meteorological raster datasets (CHIRPS precipitation) to perform lag-correlation analysis between multi-year precipitation deficits and reservoir surface shrinkage.
- **Vector Boundary Handling:** Handled vector boundaries and AOI definitions using GeoPandas and Shapely to eliminate boundary-clipping artifacts across complex, irregular reservoir branches.

Deep Learning Semantic Segmentation for Earth Observation — *Sentinel-2, PyTorch* **github.com/Saanctity/unet-water-segmentation-sentinel2**

- **Architecture Design:** Architected and trained a custom 117K-parameter U-Net semantic segmentation network from scratch in PyTorch to automate surface water extraction from satellite imagery.
- **Data Preparation:** Standardized multi-band satellite patches into analysis-ready tensors, applying data normalization, spatial tiling, and spectral feature extraction.
- **Cross-Region Validation:** Evaluated cross-region model generalization on held-out lake systems, achieving an IoU of 0.65 and F1 score of 0.79 against benchmark index masks.

Foundation-Model Fine-Tuning & SAR Hydrology Workflows — *PyTorch Lightning, TerraTorch* **Personal Lab Work**

- **EO Foundation Models:** Implemented fine-tuning pipelines using TerraTorch for Earth observation foundation models: fine-tuned TerraMind on Sen1Floods11 for dual-pol SAR flood extent mapping and Prithvi-EO-2.0 for multi-temporal land cover classification.

PROFESSIONAL EXPERIENCE

Symbioverse — *Data & Systems Analyst (Volunteer)* **Nov 2025 – Apr 2026**

- **Cloud Infrastructure:** Deployed and maintained cloud data services on Google Cloud Platform (GCP), managing Cloud SQL instances, custom DNS routing, and secure database connections.
- **Data Pipeline:** Extracted, cleaned, and normalized structured soil, agronomic, and environmental variables from agricultural scientific literature into an automated data pipeline.
- **Application Stability:** Resolved application-wide routing and accessibility bugs by refactoring layout components, ensuring dependable uptime across domains.

IISPR — *Research Intern (Machine Learning & Data)* **Jul 2025 – Oct 2025**

- **Analytical Pipelines:** Built multi-model deep learning and NLP analytical pipelines (CNN, LSTM) for classification tasks across high-volume real-world datasets.
- **Policy Insights:** Conducted structured data analysis on state-level energy datasets to model policy adoption trends and extract actionable quantitative insights.

Unified Mentor — *Machine Learning Intern* **Jun 2025 – Sep 2025**

- **Pipeline Prototyping:** Developed, benchmarked, and deployed 10 modular machine learning and computer vision pipelines across tabular, image, and spatial data.
- **QA & Validation:** Implemented quality assurance routines, automated data cleaning scripts, and structured validation workflows to optimize predictive model performance.

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech) in Computer Science & Engineering **2022 – 2026**

Radharaman University, Bhopal

- **NASA ARSET Certifications:** Fundamentals of Remote Sensing (2026); An Introduction to Synthetic Aperture Radar (SAR) and Its Applications; Monitoring Surface Water with SAR for Water Resource Management
- **WorldQuant University:** Applied Data Science Lab & Computer Vision Lab